

MICHAEL VASQUEZ

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U.S. Citizen • Clearance Eligible

PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on product development and manufacturing engineering experience across aerospace optics, aerospace lighting, and life-sciences automation. Skilled in CAD, DFM, structural FEA, and cross-functional engineering under regulated standards including FAA, UL, NEMA, and ISO 9001.

CORE COMPETENCIES

Design & Analysis: Mechanical Design, Tolerance Analysis, FEA (Static), DFM/DFA, Requirements Analysis, DOE, Gauge R&R, Root Cause Analysis

CAD/CAE Tools: SolidWorks, Creo, Fusion 360, AutoCAD, OnShape

Modeling, Simulation & Programming: MATLAB, Simulink, C++, Java

Manufacturing & Metrology: CNC (Mill/Lathe), Optical Fabrication & Assembly, Cleanroom Operations (ISO 7), Injection Molding, Sheet Metal, Additive Manufacturing (FDM/SLA), Ultrasonic Welding, Fanuc/Epson Robot Programming, Machine Vision, CMM, ShapeGrabber Ai620

Certifications: GD&T Professional (ASME Y14.5), CSWA Mechanical Design, CSWA Additive Manufacturing

PROFESSIONAL EXPERIENCE

Optical Fabrication Engineer — CACI International, *Danbury, CT* *August 2026 – Present*

- Support Aerospace optical component product lines through manufacturing operation sheets, Creo tooling/fixture design, and floor operations across CNC, optical fabrication, assembly, and cleanroom environments.
- Onboarding under paired senior-engineer mentorship toward independent product line ownership.

Mechanical Design Engineer — Point Lighting Corporation, *Bloomfield, CT* *June 2025 – August 2026*

- Owned full product lifecycle engineering for FAA-compliant aerospace lighting as sole ME.
- Designed injection-molded junction box to replace cast aluminum predecessor, reducing unit cost by ~70% while maintaining UL 50E and NEMA 4X compliance.
- Developed solar-powered FAA L-810/L-864 mounting fixture; reduced BOM cost by over 40% and expanded compatibility across multiple product lines.
- Restructured rapid prototyping workflow, cutting prototype cost by 90% while preserving FAA-compliant test fidelity.

Manufacturing Engineering Intern — ThermoFisher Scientific, *Penfield, NY* *Jan – Aug 2023, Jan – Aug 2024*

- Resolved a production-line EHS risk via a precision latch redesign; cut required force 30%, enabling single-arm use.
- Deployed automated part-presence detection systems across production lines, increasing first-pass yield by 60%.
- Reconfigured PLC controls and Epson 6-DoF robot arm sequencing, including vision system integration, on capping cell, increasing part yield by 18%.
- Engineered an automated ultrasonic welding cell for a new product assembly process, achieving 98% part yield.

ENGINEERING PROJECTS

Custom FPV Drone *Dec 2024 – Present*

- Built a MATLAB flight dynamics simulator from first principles to derive terminal velocity and impact load cases for downstream structural analysis.
- Performed FEA-driven arm optimization in SolidWorks Simulation across side, bending, and max-acceleration load cases; achieved FOS of 2 with sacrificial geometry designed to fracture outboard of the central frame, protecting avionics.

Binder-Jet Bone 3D Printer *Aug 2024 – Apr 2025*

- Led an 8-engineer team to scale an open-source binder-jet AM platform 75% for printing bone scaffolds with RevBio Tetranite bioactive adhesive.
- Designed a 2-DoF compliant Z-axis coupling and split-half piston sleeve to resolve tolerance stack-up, tune powder seal compression, and reduce tooling cost.

Three-Stage Helical Planetary Gearbox *Jan – Aug 2024*

- Designed a 27:1 planetary gearbox to motorize a standing desk; iterated spur to helical gears to reduce noise, redesigned planet carrier to resolve v1 torsional failure, and validated at ~3× FOS through load testing and dynamic analysis.

LEADERSHIP

President — RIT VEX U Robotics Team, *Rochester, NY* *Aug 2017 – May 2022*

- Led 15+ member hardware/software sub-teams to build two competition robots; won the VEX U Worlds Innovate Award.
- Served as sole CAD designer and fabricator for multiple competition robots; developed a parallel-PID controller with odometry-based localization for autonomous disturbance rejection.

EDUCATION

B.S. Mechanical Engineering — *Rochester Institute of Technology* *Aug 2020 – May 2025*